The Walk-Forward Bundle:
Specification-Curve Analysis for Trading-Strategy Validation

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Abstract

A walk-forward back-test is widely treated as the honest form of historical evaluation: parameters are fitted on a past window and evaluated on the window that follows, so no future information enters the fit. We argue that this discipline, while necessary, is not sufficient. The procedure itself requires a set of construction choices — in-sample window length, re-selection cadence, ranking objective, correlation cap — for which no principled answer exists, and each choice yields a different out-of-sample equity curve. Reporting one such curve constitutes an unreported search over construction space.

We propose reporting the *bundle*: run the full grid of defensible walk-forward constructions, retain every out-of-sample path, and report the resulting distribution rather than any element of it. This is the multiverse-analysis and specification-curve methodology of the empirical social sciences, transposed to strategy validation.

We demonstrate the procedure on two datasets processed identically. On a synthetic price series containing no edge by construction, the median of 432 walk-forward constructions is an annualized out-of-sample Sharpe of −0.27, while the best is +0.18 — a number that, reported alone, would be indistinguishable from a discovery. On S&P 500 daily data evaluated out-of-sample from 2000, the same 432 constructions give a median of +0.33 with every construction positive and a dispersion roughly half as wide. The diagnostic is not the level but the shape: the gap between the best construction and the median falls from 0.45 to 0.10. We also report the computational work required to make exhaustive evaluation cheap enough to be routine, in which a per-window memoization of the ranking metric accounted for essentially all of a 3.8× speedup. A reference implementation is released under the MIT licence.

1 The construction problem

Consider a researcher with a family of candidate strategies — parameter variants of one model, several models, or one model across several assets — and a walk-forward protocol for selecting among them. On each pass, candidates are ranked on an in-sample window, a subset is retained, and that subset is traded over the period immediately following. The out-of-sample legs are concatenated into a single track record.

Nothing in that description is unsound. The resulting equity curve contains no look-ahead: the selection at each fold uses only data preceding the period it is judged on. This is a real improvement over a single in-sample optimisation and is standard practice.

The difficulty lies one level above. Before the protocol can be run, the researcher must fix:

- the in-sample window length — 12 months? 24? 48?
- whether that window expands from a fixed origin or rolls;

- the re-selection cadence, which also fixes the length of each out-of-sample leg;
- the objective on which candidates are ranked — Sharpe, Calmar, Sortino, total return;
- how many candidates are retained, and how correlated they may be.

These are not nuisance parameters with known correct values. They are choices, and each combination produces a different out-of-sample curve. When a single curve is reported, a search over construction space has taken place and has not been disclosed — typically not even to the researcher, who experienced the process as a sequence of reasonable decisions rather than as an optimisation. The out-of-sample discipline internal to each run offers no protection, because the selection that matters occurred above it.

This is precisely the structure Gelman and Loken termed the garden of forking paths [4], and the response developed in the empirical social sciences after the replication crisis is instructive: rather than defend one specification, report all of them. Steegen et al. [7] call this a multiverse analysis; Simonsohn et al. [6] formalise the reporting as a specification curve. In finance the closely related concern — that the reported performance of a strategy reflects the number of trials that produced it — is treated by White's reality check [8], by the multiple-testing haircut of Harvey et al. [5], and by the backtest-overfitting programme of Bailey et al. [2, 3]. Our contribution is neither a new statistic nor a new correction. It is the observation that the walk-forward protocol has its own multiverse, that this multiverse is small enough to enumerate exhaustively, and that its shape is directly interpretable.

2 Method

2.1 Setup

Let $R \in \mathbb{R}^{T\times k}$ be a matrix of candidate return streams, column j being the realised P&L of candidate j at time t, and let $V \in \mathbb{R}^{T\times k}$ be the corresponding traded notional (turnover). We require that R be *causal*: each column must be the output of a rule whose parameters were not fitted using data after t. Nothing downstream can repair a violation of this requirement.

A construction c is a tuple

$$c = (L,\ s,\ a,\ m,\ n,\ \rho,\ K),$$

with L the in-sample window length in months, s the re-selection cadence (equal to the out-of-sample leg length), $a \in \{\text{expanding}, \text{rolling}\}$, m the ranking objective, n the shortlist size, ρ the correlation cap and K the maximum number of candidates deployed.

2.2 One construction

Given a first out-of-sample date t_0, the schedule generates windows $[\,u_i, v_i\,]$ with $v_i = t_0 + i\cdot s$ and $u_i = v_0 - L$ when a is expanding, $u_i = v_i - L$ when rolling. The schedule terminates s months before the end of the sample, so that every selection is followed by a period on which it can be judged.

For each window, in order:

1. **Rank.** Evaluate m on $R[u_i{:}v_i]$ column-wise and retain the best n candidates.
2. **De-correlate.** Traverse the shortlist best-first, admitting a candidate only if its correlation with every already-admitted candidate is below ρ; stop at K admissions. This is deliberately greedy and deterministic: explicability matters more than optimality when the object of the exercise is a robustness claim.

3. **Gate on economics (optional).** Discard candidates whose realised P&L per unit of turnover, $\sum_t R_{tj} / \sum_t V_{tj}$, falls below a floor. This is the quantity that must exceed transaction cost.
4. **Evaluate.** Equal-weight the survivors over $(v_i,\, v_i + s]$ and record the resulting return and turnover series.

The out-of-sample leg begins strictly after the in-sample window closes; the two never share an observation. Concatenating the legs yields a single continuous out-of-sample return stream $r^{(c)}$, from which we compute an annualized Sharpe ratio $\widehat{S}(c)$.

2.3 The bundle

Let $\mathcal{C}$ be the Cartesian product of the admissible values of each construction parameter. The object we report is the collection

$$\mathcal{B} \;=\; \{\, r^{(c)} \;:\; c \in \mathcal{C} \,\},$$

together with the induced distribution of $\widehat{S}(c)$ over $\mathcal{C}$. The summary statistics of interest are the median, the dispersion, the fraction of constructions with positive out-of-sample Sharpe, and — the quantity we find most diagnostic — the gap

$$\Delta \;=\; \max_{c \in \mathcal{C}} \widehat{S}(c) \;-\; \mathrm{med}_{c \in \mathcal{C}}\, \widehat{S}(c),$$

which measures how much apparent performance is available to a researcher who selects a construction after seeing the results.

A grid should span choices the researcher genuinely cannot rule out, and no others. Adding axes on which one has no opinion inflates $|\mathcal{C}|$ without adding information; adding axes on which one setting is known to be wrong imports a known-bad configuration into the reported distribution.

3 Two experiments

Both experiments use an identical pipeline: a sweep of 105 trend-following rules (Donchian breakout, exponential-moving-average crossover, and time-series momentum, across six fast and six slow look-backs), volatility-targeted, charged 2 basis points per unit of traded notional, executed one bar after signal. The construction grid is $L \in \{12, 24, 36, 48\}$, $s \in \{3, 6, 12\}$, $a \in \{\text{expanding}, \text{rolling}\}$, $m \in \{\text{Sharpe}, \text{Calmar}, \text{Sortino}\}$, $\rho \in \{0.3, 0.5, 0.7\}$, $K \in \{5, 10\}$, with $n = 20$ throughout — $|\mathcal{C}| = 432$.

3.1 A synthetic null

The first price series is generated: a volatility-clustering, fat-tailed process with constant drift. It has the texture of a market — long quiet stretches punctuated by violent moves, so trend rules behave as they do on real data — and, by construction, contains nothing a look-back rule can predict. The individual candidates have a median annualized Sharpe of -0.09: after costs, the family loses money, as it should.

Figure 1 shows all 432 out-of-sample paths. The median construction returns an annualized out-of-sample Sharpe of -0.27 and only 5% of constructions end positive. The best returns $+0.18$.

That $+0.18$ deserves emphasis. It is arithmetically correct, strictly out-of-sample, produced by a defensible construction, and entirely an artefact of having tried 432 of them on data

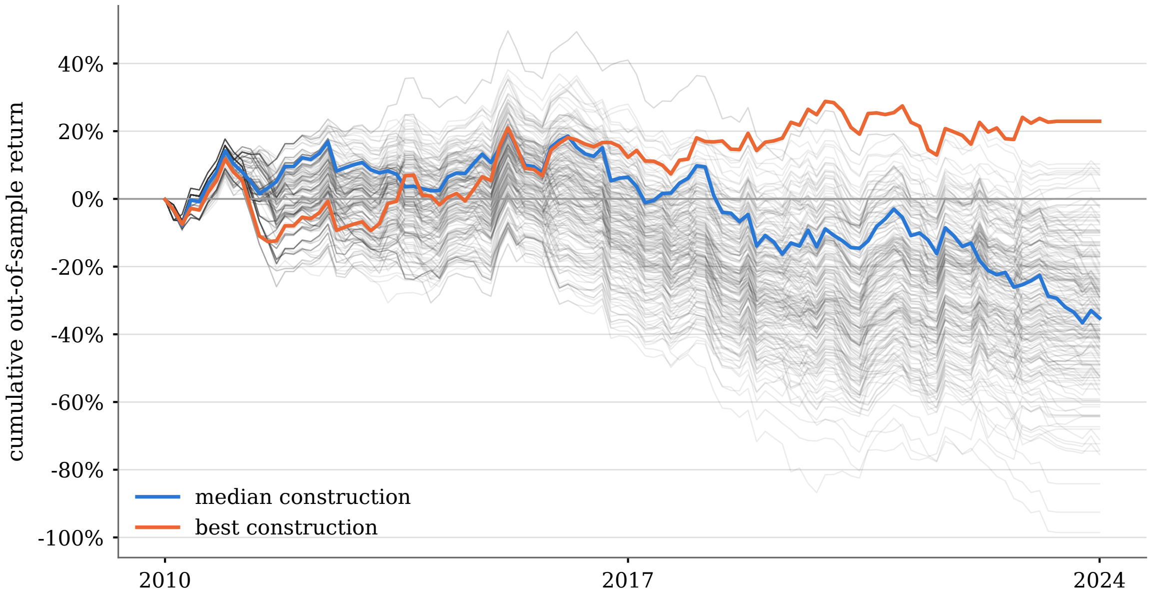

Figure 1: All 432 walk-forward constructions on the synthetic null. Each faint line is one construction's concatenated out-of-sample return stream. Every line is a genuine walk-forward result: fitted on a past window, traded on the window after it, no look-ahead. The data contains no edge.

Table 1: The bundle on both datasets. Annualized out-of-sample Sharpe over 432 walk-forward constructions.

	Synthetic	**S&P 500**
Median	−0.27	+0.33
Minimum	−0.79	+0.06
Maximum	+0.18	+0.43
Dispersion (s.d.)	0.16	0.09
Constructions with $\widehat{S} > 0$	5%	100%
Δ (best − median)	0.45	0.10

engineered to contain nothing. A researcher who ran this grid, selected the best construction and reported its curve would not be committing fraud; they would be doing what the field's reporting conventions permit.

3.2 S&P 500

The second series is the S&P 500 daily index from 1927, with the first out-of-sample date set to January 2000. The same 105 rules and the same 432 constructions are applied without modification. Individual candidates have a median annualized Sharpe of 0.45.

3.3 What separates them

A median out-of-sample Sharpe of 0.33 for plain trend following on an equity index is unremarkable, and should be: it is approximately the modest, well-documented figure the literature would lead one to expect. The level is not the finding.

The finding is the shape. Dispersion falls from 0.16 to 0.09. The fraction of positive constructions rises from 5% to 100%. And Δ falls from 0.45 to 0.10: on the synthetic null, choosing the best

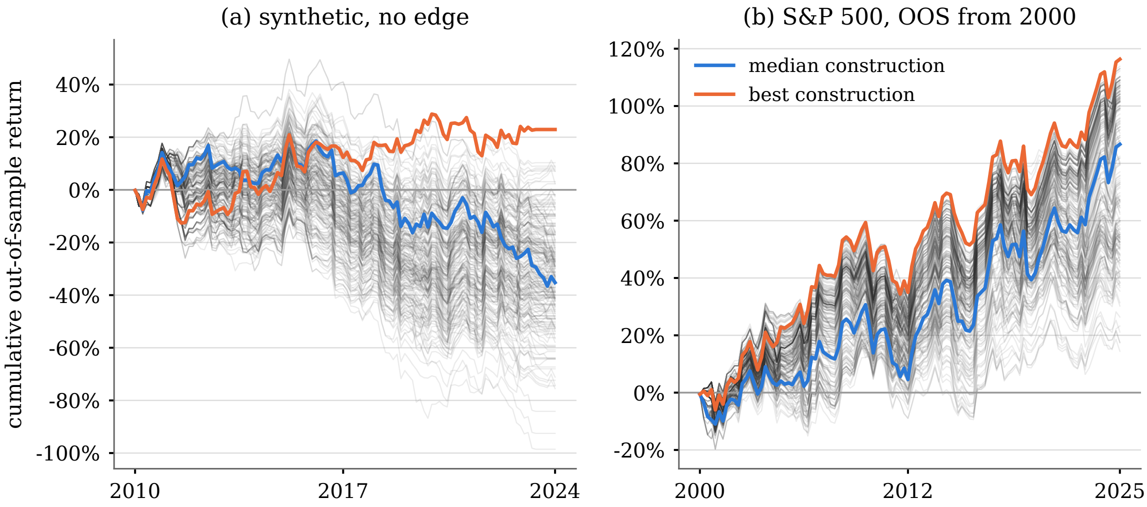

Figure 2: The same procedure on both datasets. (a) The synthetic null: a wide fan, mostly below zero. (b) S&P 500 out-of-sample from 2000: a tight bundle, entirely above zero. The vertical scales differ; the comparison of interest is the width of each fan relative to its own level, quantified in Table 1.

construction after the fact buys 0.45 of Sharpe that does not exist, whereas on the S&P it buys 0.10, because there is far less to choose *between*.

Figure 3 shows the two distributions directly. This is the specification curve of [6] in histogram form, and it is what we propose reporting in place of a single equity curve.

3.4 A caveat on reading the fans directly

Figure 2 is drawn in cumulative-return space, where the visual width of a fan is not a measure of construction sensitivity. The S&P bundle is in fact *wider* in raw terms — the 10–90% band spans 65.6 percentage points at the terminal date against 50.2 for the synthetic null — simply because it is compounding a positive return. Sensitivity is a ratio, not a distance.

Figure 4 therefore plots each path as a deviation from its bundle's median path, expressed in multiples of that bundle's median terminal level. On that scale the terminal 10–90% band is 1.55× the level for the synthetic null and 0.78× for the S&P: twice as tight. This is the representation we recommend for visual reporting, and it agrees with the Sharpe dispersion in Table 1, as it must.

3.5 How large must the grid be?

If reporting a distribution over constructions is the remedy, the choice of which constructions to enumerate is itself a researcher degree of freedom, and the objection of infinite regress is fair. It does not, however, bite in practice, for three reasons.

First, the axes are the protocol's own parameters, not the model's. They are few, they are enumerable, and each admits a handful of defensible values — unlike a model's parameter space, which is unbounded.

Second, the response surface over those axes is smooth. Inserting a fifth in-sample window length between 24 and 36 months tells one almost nothing that 24 and 36 did not.

Third, and decisively, the required grid size is an empirical question with a measurable answer. Table 2 reports the accuracy with which a random subgrid of size n recovers the median of the

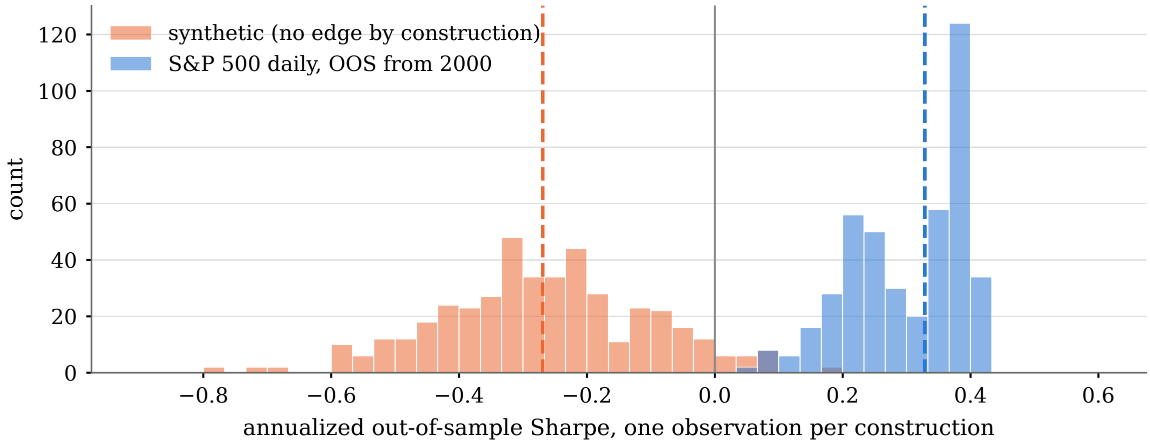

Figure 3: Distribution of annualized out-of-sample Sharpe across the 432 constructions, one observation per construction. Dashed lines mark the medians. The separation between the two distributions — in location, in width, and in support — is the diagnostic.

full 432-point grid.

Table 2: Mean absolute error in the estimated grid median, over 4,000 random subgrids of each size, against the full 432-point grid.

Subgrid size n	Synthetic	S&P 500
5	0.066	0.054
10	0.045	0.040
20	0.032	0.034
50	0.020	0.025
100	0.013	0.019
200	0.007	0.013

The grid needs to be large enough that the estimation error is small relative to the effect being examined, and no larger. The difference under test here — a median of +0.33 against −0.27, a gap of 0.60 — is roughly eighteen times the estimation error at $n = 20$. Enumerating all 432 buys precision that the decision does not consume. We report the full grid because it is cheap (Section 5), not because the conclusion requires it.

The regress terminates, then, not because the grid choice is objective, but because it is *declared*. A single curve conceals its search; a bundle reported with its grid exposes the remaining discretion to the reader, who can object to a specific omitted axis. That is a weaker guarantee than objectivity and a much stronger one than convention.

3.6 Marginal effects

Because the grid is exhaustive, the marginal effect of each construction choice is available by conditioning. On the S&P data three of the four principal axes are flat: median Sharpe varies by less than 0.01 across in-sample window lengths (0.324–0.332) and by 0.01 across ranking objectives.

One axis is not flat. Rolling windows give a median of 0.23; expanding windows give 0.38. A researcher who adopted an expanding window as a default, reported the curve and never swept the alternative would have taken an unmeasured 0.15 Sharpe difference on the strength of an unexamined convention. Notably, the same axis has the opposite and much weaker sign on the

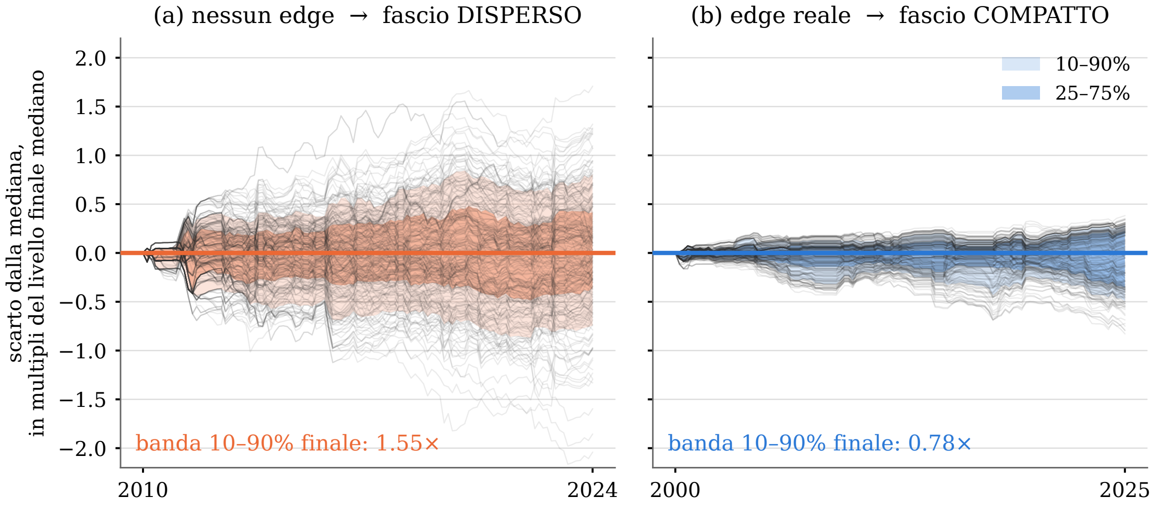

Figure 4: Construction sensitivity on a common scale. Each faint line is one construction's deviation from the median construction, divided by the bundle's median terminal level. Shaded bands are the inter-decile and inter-quartile ranges. A wide fan (a) means the reported result depends on which construction was chosen; a tight one (b) means it does not.

synthetic null (-0.25 versus -0.30), which is what one expects of an effect that interacts with the presence of real signal rather than reflecting a universal preference.

This is the practical case for the bundle, independent of the epistemic one: it converts every unexamined default into a measurement.

4 Making exhaustive evaluation cheap

Exhaustive enumeration is a methodological prescription only if the enumeration is affordable. We report the optimisation work because its lesson generalises and because the naive result was misleading.

The workload has an unusual structure: it is *redundant* before it is *parallel*. Every construction traverses the same candidate matrix, and constructions differing only in ρ or K produce identical in-sample rankings, since those parameters act after the ranking.

Instrumented timings on a grid of 576 constructions over a 5000×1000 candidate matrix attributed 82.9% of runtime to a single operation: the in-sample ranking objective, recomputed over the full window slice once per window per construction. Out-of-sample leg assembly accounted for 10.0%, greedy de-correlation for 2.9%, and slicing for 1.2%. By Amdahl's law [1], no amount of work on the remaining 17.1% could have produced more than a 1.2× speedup — a ceiling we established only after implementing two optimisations that fell below it.

The effective change was a memoization of the ranking objective keyed on (objective, window), shared across constructions within a process. On the benchmark grid this reduced 5,952 evaluations to 656, and the predicted and observed runtimes agree closely:

$$45.3\,\mathrm{s} \times \left(0.171 + \tfrac{0.829}{9.1}\right) = 11.9\,\mathrm{s} \quad \text{predicted,} \qquad 12.0\,\mathrm{s} \text{ measured.}$$

Two results are worth recording for practitioners. First, a closed-form prefix-sum evaluation of moment-based objectives — asymptotically the strongest optimisation available, reducing per-window cost from $O(Tk)$ to $O(k)$ — yielded no measurable gain on a wide grid, because the memoization had already eliminated the repetition it exploits; it returns a 1.5× gain only in the regime where each schedule appears once. Second, the obvious parallelisation, dispatching one construction per worker, was 5.9× *slower* than serial execution: each task shipped a fresh copy of the candidate matrix and began with an empty cache, so the eliminated work returned, multiplied by the number of tasks. Grouping constructions that share a schedule into the same task restores the gain. Parallelism that destroys locality is a regression.

5 Limitations

The procedure inherits the causality of its inputs and cannot verify it. If a candidate's P&L was generated by a rule fitted on the full history, every number reported here is contaminated regardless of the walk-forward discipline applied on top.

A tight bundle establishes insensitivity to construction choice. It does not establish that the edge will persist, that the candidate universe was not itself the product of an unreported search, or that the strategy is implementable at size. It answers one question well and is silent on the others.

The dispersion of $\widehat{S}$ over $\mathcal{C}$ is not a sampling distribution and admits no direct frequentist interpretation; constructions share data and are strongly dependent. We report it as a descriptive summary of sensitivity, not as an interval estimate. Formal inference over specification spaces — as developed in [6] — is compatible with this procedure but is not attempted here.

Finally, the grid is a modelling choice like any other, and a researcher determined to produce a tight bundle can do so by narrowing it. The grid must be reported in full alongside the result, and we regard a bundle reported without its grid as no more informative than a single curve.

6 Availability

A reference implementation is released under the MIT licence at

https://github.com/epiphanyalpha/walkforward-bundle

Both experiments in this paper reproduce from a single command, with no data download:

```
pip install walkforward-bundle
python -m validation.demo
```

The synthetic series is generated deterministically from a fixed seed; the S&P 500 results are reproduced by passing a daily close series via `--csv`. No market data is redistributed with the package.

References

[1] Amdahl, G. M. (1967). Validity of the single processor approach to achieving large scale computing capabilities. *AFIPS Conference Proceedings*, 30, 483–485.

[2] Bailey, D. H., & López de Prado, M. (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality. *Journal of Portfolio Management*, 40(5), 94–107.

[3] Bailey, D. H., Borwein, J. M., López de Prado, M., & Zhu, Q. J. (2015). The Probability of Backtest Overfitting. Working paper, revised February 2015.

[4] Gelman, A., & Loken, E. (2013). The garden of forking paths: Why multiple comparisons can be a problem, even when there is no "fishing expedition". Working paper, Department of Statistics, Columbia University.

[5] Harvey, C. R., Liu, Y., & Zhu, H. (2016). ... and the Cross-Section of Expected Returns. *Review of Financial Studies*, 29(1), 5–68.

[6] Simonsohn, U., Simmons, J. P., & Nelson, L. D. (2020). Specification curve analysis. *Nature Human Behaviour*, 4, 1208–1214.

[7] Steegen, S., Tuerlinckx, F., Gelman, A., & Vanpaemel, W. (2016). Increasing Transparency Through a Multiverse Analysis. *Perspectives on Psychological Science*, 11(5), 702–712.

[8] White, H. (2000). A Reality Check for Data Snooping. *Econometrica*, 68(5), 1097–1126.